MUHAMMAD ZAIN UL ABEDIN

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Hiring Team
Abyss Solutions

Application - Visual Inspection and Condition Assessment

Dear Hiring Team,

I am applying for the visual inspection and condition assessment role at Abyss Solutions. I completed my Mechanical Engineering degree at International Islamic University Islamabad in 2019 with a CGPA of 3.57/4.00. Your use of computer vision and 3D digital twins to guide infrastructure maintenance matches my interest in practical engineering data analysis.

For my independent portfolio project, AssetInspect, I developed a workflow for reviewing synthetic pipeline, tank and marine-structure imagery. It supports manual data annotation of faults and defects, identification of mechanical, structural and hydraulic features, equipment tags, and documented observations with recommended follow-up. Raster P&IDs, isometrics, schematics and CAD screenshots can also be annotated, giving each record clear visual context.

I incorporated inspector and reviewer QA checklists, review comments, change requests, resubmission and traceable audit history. Assessment records link preliminary findings with client-defined criteria, rationale, reference documents and scope limitations. Engineering inspection reports and Excel-compatible CSV/XML exports help communicate evidence and recommended actions. This work reinforced the importance of distinguishing visible observations from conclusions requiring further engineering verification.

AssetInspect also supports 3D model and point cloud review using OBJ, ASCII PLY and XYZ/CSV data, with two-point measurements in declared model coordinates. Alongside this project work, I have familiarity with Autodesk AutoCAD, Navisworks and ReCap; MS Excel, Google Sheets and Linux; and API, NACE, ASCE and ASME inspection and coating standards. I am keen to extend this foundation to your photogrammetric modelling and metrology workflows.

I would bring attention to detail, analytical problem solving, clear written communication and a collaborative approach to review feedback. I welcome the opportunity to demonstrate AssetInspect and discuss how I could support accurate visual data analysis, consistent quality assurance and actionable condition assessment at Abyss Solutions. Thank you for considering my application.

Yours sincerely,

Muhammad Zain Ul Abedin